

Class test discussion

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→ 1) FOR ALL CONSTANTS $c > 0$, $\log n = O(n^c)$

$$2) T(n) = \underbrace{8}_a T\left(\underbrace{\frac{n}{5}}_b\right) + 7n^c$$

$$\log_b a \text{ vs } 1 \quad \log_{\frac{1}{5}} 8 < \log_{\frac{1}{5}} 5 < \log_{\frac{1}{5}} 25 \text{ vs } 1$$

$$T(n) = \Theta(n^{\log_5 8})$$

$$3) T(n) = \underline{100} T\left(\underline{\frac{n}{10}}\right) + 1000 n^{10}$$

$$\log_{\frac{1}{10}} 100 \text{ vs } \underline{10}$$

$$T(n) = \Theta(n^{10})$$

$$4) T(n) = 5T\left(\frac{n}{5}\right) + 5n^c \log^5 n$$

$$1 = \log_5 5 \text{ vs } 1$$

$$T(n) = \Theta(f(n) \cdot \log n) \\ = \Theta(n \log^6 n)$$

5) KARATSUBA'S FAST INTEGER MULT. ALG.

MULT. 2 n -BIT NUMBERS $\longrightarrow$ 3 SUB-PROBLEMS

MULT 2 $\frac{n}{2}$ -BIT NUMBERS EACH.

6) Repeatedly toss a fair coin until it lands Heads, then stop
Repeat experiment twice. X = #tosses in first expt.,
 Y = #tosses in second expt.

What is $E[X+Y]$?

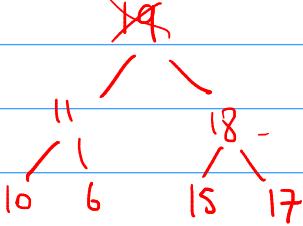
$$\text{LIN. EXP: } E[X+Y] = \overbrace{E[X]}^2 + \overbrace{E[Y]}^2 = 4$$

7) A bag contains 100 balls numbered 1 to 100. What is the probability that a randomly selected ball has a number between 1 and 40?

$$\frac{40}{100} = \frac{2}{5}$$

8) EXPECTED RUNNING TIME OF RAMP. SELECT TO FIND $n/2$ 'TH LARGEST ELT. : $O(n)$.

9)



ROOT VALUE AFTER 4 REMOVE MAX OPS.?

19 18 17 15 10 6

10) RUNNING TIME OF REMOVE MAX : $O(\log n)$

11) COMARISON-BASED SORTING ALGS: HEAPSORT, MERGESORT, QUICKSORT
NON COMP.-BASED ALGS : RADIX SORT, COUNTING SORT

12) ALL COMP.-BASED SORTING ALGS MUST TAKE $\Omega(n \log n)$ TIME (WORST-CASE).

13) # BITS REQUIRED TO WRITE n 'TH FIB. NUMBER?
 $F(n) \approx 1.6^n$ # BITS = $O(\log 1.6^n)$
 $= n \cdot \underbrace{O(\log 1.6)}_{\text{CONST.}} = O(n)$

14) THERE IS A DP ALG FOR THE KNAPSACK PROBLEM,
RUNNING TIME $= O(nW)$

15) INTERVAL SCHEDULING GREEDY ALG: ALWAYS PICK NEXT VALID INTERVAL THAT FINISHES THE EARLIEST.